

Nardiena A. Pratama

Linkedin

Edinburgh, UK

Github

Education

- University of Edinburgh, UK** Sept 2025 - Present
Centre for Doctoral Training (CDT) in Artificial Intelligence for Biomedical Innovation
Supervisors: Prof. Ajitha Rajan and Prof. Paul Brennan
Project: Explainable AI Models for Glioblastoma Treatment Using Brain MRI
- The University of Queensland, Australia** Feb 2023-Nov 2023
Bachelor of Computer Science (Honours) awarded with First Class Honours
Thesis: *MRI Segmentation in the Latent Space* with Dr. Shekhar Chandra
 - Completed Honours while working 3 days/week as a Data Scientist
- The University of Queensland, Australia** Feb 2020-Nov 2021
Bachelor of Computer Science (Data Science Major) - Joint Double Degree Program
- University of Indonesia** Aug 2017-Dec 2019
Bachelor of Computer Science - Joint Double Degree Program

In Press

- S. Civelli, P. Bernardelle, **N. A. Pratama**, G. Demartini. *Ideology-Based LLMs for Content Moderation*. [Full Paper]. *ACM Transactions on Intelligent Systems and Technology*, 2026. [Link to Paper].
- N. A. Pratama**, S. Fan, G. Demartini. *Visual Content Annotation: Differences between Humans and Foundation Models*. [Full Paper]. *AAAI Conference on Web and Social Media (ICWSM '25)*. [Link to Paper].
- E. Meguellati, **N. Pratama**, S. Sadiq, G. Demartini. *Are Large Language Models Good Data Preprocessors?* [Short Paper]. *WWW '25: The ACM Web Conference 2025*. [Link to Paper].
- M. Skipanes, **N. A. Pratama**, K. Porter, G. Demartini. *Fast Synthetic Data Generation for Case-Specific Entity Extraction in Criminal Investigations*. [Full Paper]. *Digital Forensics Doctoral Symposium (DFDS '25)*. [Link to Paper].

Research Experience

- UQ Casual Research Assistant** with Prof. Gianluca Demartini Dec 2023-July 2025
Working on individual project and assisting with PhD projects.
 - Extending the *Human vs Machine Learning Data Annotations on Visual Content* Project to include ML-generated captions in the analysis; improving the training process of the predictive models
 - Assisted a PhD student by contributing to code development for a Named Entity Recognition (NER) pipeline using a BERT-based transformer model to use with Norwegian Police documents
 - Assisting a PhD student with generating and cleaning image captions using captioning models and Large Language Models (LLMs) for a project investigating persuasive techniques using LLMs
- UQ Honours Thesis** with Dr. Shekhar Chandra Feb 2023-Nov 2023
Worked on *MRI Segmentation in the Latent Space Using Deep Learning*.
 - Developed a deep learning pipeline, consisting of pretrained Vector Quantised Variational Autoencoders and a U-Net-like model, to perform segmentation and reconstruction of 2D MRI data.
- UQ Semester-long Research Project** with Prof. Gianluca Demartini July 2023-Nov 2023
Worked on *Human vs Machine Learning Data Annotations on Visual Content* project.
 - Compared and analysed human annotations and machine learning (ML) generated annotations for handwashing data using statistical techniques
 - Developed and analysed predictive machine learning models using human and ML annotations
- UQ Winter Research Program** with Prof. Gianluca Demartini June 2023-July 2023
Worked on *Human vs Machine Learning Data Annotations on Visual Content* project.
- Telstra Labs Software Engineering Intern** with Luca Stamatescu Nov 2020-Feb 2021
Worked on *Synthetic Dataset Generation* project at Telstra Labs.
 - Created a synthetic dataset generator which creates significant amounts of artificial images of the Telstra modem using the Unity software, cutting down weeks' worth of collecting training data to several minutes

- Developed a transfer learning-based Faster-RCNN object detection model using Tensorflow's Object Detection API, trained using synthetic data of the Telstra modem generated from the synthetic dataset generator

Presentations

Research Spotlight Competition: Interpretable AI for Glioblastoma Treatment Prediction Using Brain MRI. Finalist & People's Choice Award Winner. **London Hopper Colloquium, London, UK**, May 21, 2026.

3-Minute Thesis Presentation: Enhancing Breast Cancer Classification with LLM-Generated Textual Descriptions. Runner-Up Award. **Information Resilience PhD School 2024**, Oct 2024.

MRI Segmentation in the Latent Space Using Deep Learning. **UQ EECS Innovation Showcase 2023**, Nov 2023.

Evaluating the Human and Artificial Intelligence Perception of Digital Media for Socio-Economic Bias. **Australasian Conference for Undergraduate Research (ACUR) 2023**, Nov 2023.

Industry Experience

Telstra Australia

Feb 2022-August 2025

Graduate Data Specialist

Feb 2022-Mar 2023

- Contributed to data exploration within the Telstra and Quantum Joint Venture to identify key features for developing fraud prevention machine learning models in the Australian context
- Designed and developed mathematical optimisation model using PuLP to identify optimal edge compute locations across Australia under complex stakeholder constraints, integrating it into a larger network-planning system.

Emerging Technology Specialist (Data Scientist)

April 2023-August 2025

- Conducted trials for vehicle detection, analysing data from underground fibre optic networks using object detection and image segmentation models
- Developed a high-performance data pipeline using Polars to process large-scale network data and compute metrics for downstream analysis
- Designed a scalable and reusable database schema and test framework to integrate multi-source enterprise data, simplifying long-term data management

Technical Skills

- Programming Languages: Python, Java, C#, Javascript, React Native, SQL
- Libraries/frameworks: Pandas, Polars, Tensorflow, Pytorch, OpenCV, Darknet
- Deployment: Tensorflow Serving, Docker
- Other: Bash scripting, CUDA, Git version control

Course Projects

CSSE 6400: Salt & Paper - Recipe Web Application

Feb 2023-June 2023

- Led and collaborated with a team of four students, including undergraduates and postgraduates, to develop a web application that allows users to create, view, and share recipes in the platform
- Designed and implemented the software architecture, including containerisation of the load balancer, Celery workers, Redis, and PostgreSQL database using Docker Compose
- Orchestrated the deployment process with Terraform on Amazon Web Services (AWS), utilising Amazon ECS for task definition and Amazon SQS for efficient queue management

DECO 3801: Plateducate - AI Diet Mobile Application

July 2021-Nov 2021

- Partnered with five undergraduate students and worked on a mobile application that tracks user's food intake to improve their diet and avoid health problems
- Developed YOLO-based food object detection model using Darknet for automatic food identification to enhance user experience
- Containerised the backend and ML applications within the mobile app using Docker Compose, and implemented TensorFlow Serving to facilitate model serving

COSC 3000: Roller Coaster Simulation

Feb 2020-June 2020

- Developed a roller coaster simulation in Unity with C# scripts, featuring online assets, camera controls (1st/3rd person POV, pan), and movement controls (start/stop car).
- Aimed to mimic the experience of riding an actual roller coaster, focusing on a small track section for immersive testing.

Other Experience

University of Indonesia Linear Algebra, *Teaching Assistant*

Feb 2019-June 2019

University of Indonesia Formal Languages and Automata, *Teaching Assistant*

Aug 2019-Dec 2019